

CCNA 200-301 V2.0

06 TCP, UDP and the Diagnostic Toolkit

Part I — Foundations

EXAM TOPICS

2.4

LECTURER

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What you will be able to do

- **Explain** what a port number identifies, and recognise the well-known ports that appear in access lists and in fault reports.
- **Choose** correctly between TCP and UDP behaviour when reasoning about a symptom.
- **Use** ping, extended ping and traceroute deliberately, varying source, size and count to test a specific hypothesis.
- **Read** packet capture output well enough to say which direction failed.

Before we start

Chapter 4 for addressing, Chapter 3 for what a switch does with a frame.

Vocabulary for this chapter

port

socket

TCP

UDP

three-way handshake

acknowledgement

retransmission

well-known port

ephemeral port

ICMP

echo request

TTL

extended ping

traceroute

Each is defined where it first matters — not here.

The toolkit is the exam

This is what the blueprint actually asks for

Topic **2.4** names the tools by name: “show commands (including show logs), ping, extended ping, trace route, and packet capture output”. It is one of the eight *troubleshoot* topics, and the tools in it are how you will answer the other seven. Learn them as instruments, not as trivia.

TCP or UDP, and what each failure looks like

TCP establishes a connection with a three-way handshake — SYN, SYN-ACK, ACK — numbers every byte, acknowledges what arrives and retransmits what does not. It costs latency and gains reliability.

UDP sends and forgets. No handshake, no acknowledgement, no retransmission. It is right for anything where a late packet is worse than a lost one, and for short exchanges where a handshake would double the cost — which is why DNS and DHCP use it.

The diagnostic value is in how they fail. A blocked TCP port produces a *connection timeout or refusal* — something definite. A blocked UDP port often produces *nothing at all*, because there was never going to be an acknowledgement. “It just hangs” points at UDP or at a one-way path; “connection refused” means something answered and said no.

ping 192.168.20.10

```
R1# ping 192.168.20.10
```

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.20.10, timeout is 2 seconds:
```

```
!!!!
```

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
```

The extended ping is the one that finds faults

A plain ping from a router sources the packet from the exit interface. That is often not the address you care about. **Extended ping** lets you set the source, and this single option answers the commonest routing question: “does the far end have a route *back* to this subnet?”

Run `ping` with no arguments to enter the extended dialogue.

ping

R1# ping

```
Protocol [ip]:
Target IP address: 192.168.30.10
Repeat count [5]: 100
Datagram size [100]: 1500
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 192.168.20.1
Type of service [0]:
Set DF bit in IP header? [no]: yes
Sweep range of sizes [n]:
Sending 100, 1500-byte ICMP Echos to 192.168.30.10, timeout is 2 seconds:
Packet sent with a source address of 192.168.20.1
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
Success rate is 100 percent (100/100), round-trip min/avg/max = 1/3/12 ms
```

traceroute 192.168.30.10

```
R1# traceroute 192.168.30.10
```

```
Type escape sequence to abort.
```

```
Tracing the route to 192.168.30.10
```

```
 1 10.0.12.2 4 msec 1 msec 1 msec
 2 10.0.23.3 8 msec 5 msec 4 msec
 3 * * *
 4 * * *
```

Where a traceroute stops is not always where the fault is

Asterisks mean no reply from that hop. That can mean the fault is there — or that the hop is configured not to reply, or that the *return* path from that hop is broken while the forward path is fine. A traceroute that dies at hop 3 narrows the search to hop 3 onwards; it does not convict hop 3. Confirm with an extended ping sourced from an address the far end should know.

show capture BUF

```
SW1# show capture BUF
```

```
1  0.000000  192.168.20.10 -> 192.168.30.10  ICMP Echo request  id=1 seq=1
2  2.001113  192.168.20.10 -> 192.168.30.10  ICMP Echo request  id=1 seq=2
3  4.002240  192.168.20.10 -> 192.168.30.10  ICMP Echo request  id=1 seq=3
```

Reading a capture

- **Treating a failed ping as proof of a broken forward path.** Ping needs both directions. Half the time the forward path is fine.
- **Pinging from the wrong source.** A router pinging from its exit interface tests a path the users never take.
- **Concluding from five packets.** Intermittent loss needs a hundred.
- **Forgetting that ICMP may be filtered.** Plenty of hosts and firewalls drop echo requests by policy. "It does not ping" is not the same as "it is down".

Web server reachable by IP, refuses connections by name and port.

Users cannot load an internal site. From a workstation, `ping 10.20.5.40` succeeds with 0% loss. Browsing to the address times out.

Web server reachable by IP, refuses connections by name and port.

What would you check first — and what do you expect to see?

Why it is doing that

Diagnosis. Ping proves layer 3 connectivity in both directions, so addressing, masks, routing and the physical path are all sound. The failure is above layer 3 — either the service is not listening, or something is filtering TCP 443 specifically. A timeout rather than a refusal points at a filter: a closed port on a live host normally answers with a reset.

And the lesson

Fix. Test the port rather than the host. From a router on the path, an extended ping cannot help — it is ICMP only — so use `telnet 10.20.5.40 443`: a blank screen means the port is open, a refusal means closed, and a hang means filtered. Then walk the access lists on the path (Chapter 24). Do not touch addressing; the ping already cleared it.

Diagnose with intent

Topology. Two routers back to back, one LAN each, one PC per LAN.

1. Address everything and confirm end-to-end ping between the PCs.
2. From **R1**, run a plain ping to the far PC, then an extended ping sourced from the LAN interface. Explain why the second is the more useful test.
3. Remove the return route on **R2**. Predict what a ping from PC-A will show, and what a traceroute will show, *before* testing.
4. Restore the route. Now set the far LAN interface MTU to 1400, and use an extended ping with size 1500 and the DF bit set. Record the result.
5. Run a hundred-packet ping and record whether any loss appears.

Diagnose with intent (continued)

1. Extension. Have a partner break one thing — an address, a mask, a route or an interface — and identify which, using only ping, extended ping and traceroute, in under five minutes.

CHECKPOINT 1 of 3

A ping returns U.U.U. What has happened, and why is that more useful than five dots?

Discuss · then advance

Checkpoint 1 of 3

A ping returns U.U.U. What has happened, and why is that more useful than five dots?

U is an ICMP destination-unreachable reply. Something on the path actively told you it could not deliver — so the packet was routed, and a device took responsibility. Five dots means silence, which could be anything from a dropped packet to an ACL to a dead host. A reply is always more informative than a timeout.

CHECKPOINT 2 of 3

Why does an extended ping with a specified source often succeed when a plain ping from the same router fails?

Discuss · then advance

Checkpoint 2 of 3

Why does an extended ping with a specified source often succeed when a plain ping from the same router fails?

Because a plain ping is sourced from the exit interface's address, which the far end may have no route back to. Sourcing from a LAN interface or loopback uses an address the far end's routing table knows about.

CHECKPOINT 3 of 3

A traceroute shows three hops then asterisks. List three different explanations.

Discuss · then advance

Checkpoint 3 of 3

A traceroute shows three hops then asterisks. List three different explanations.

Any of: an ACL blocking the probes; a device configured not to send ICMP time-exceeded messages; asymmetric return path with no route back; a firewall dropping the traffic; or the destination genuinely unreachable beyond that hop.

What to take away

- A port identifies the program; address plus port is a socket.
- TCP handshakes, acknowledges and retransmits; UDP does none of it. Their failure modes differ, and the difference is diagnostic.
- Ping tests both directions at once. A success rules out a great deal; a failure convicts nothing on its own.
- Extended ping sets the source, size and DF bit — it is the tool that finds return-path and MTU faults.
- Traceroute narrows the search; it does not name the culprit.
- A capture showing requests and no replies is a one-way path.

Where we got to

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- Extended ping sets the source, size and DF bit — it is the tool that finds return-path and MTU faults.

NEXT

Chapter 7 — The IOS Command Line